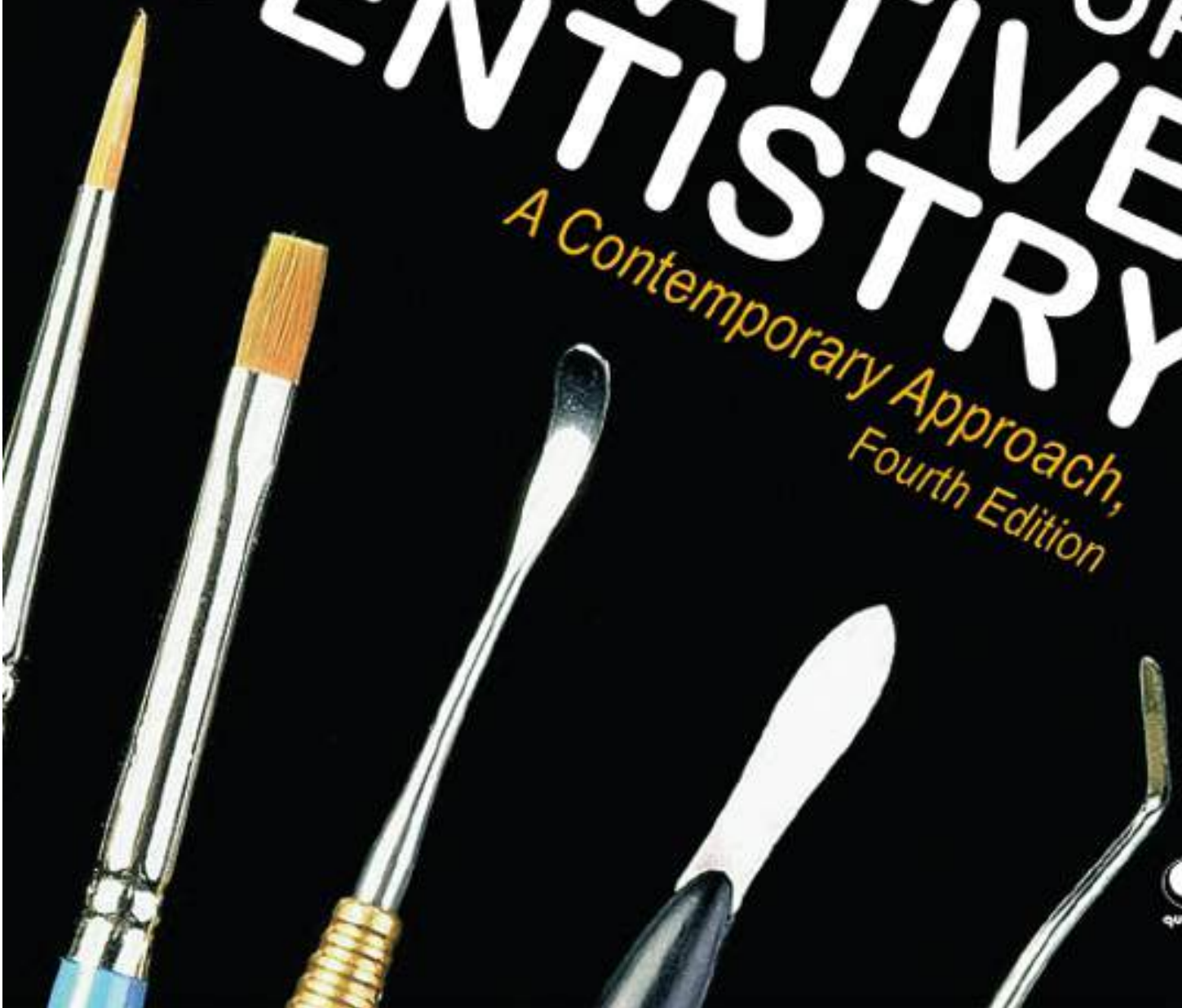
SUMMITT'S FUNDAMENTALS OF OPERATIVE DENTISTRY

A Contemporary Approach, Fourth Edition

Edited by
Thomas J. Hilton, DMD, MS
Jack L. Ferracane, PhD
James C. Broome, DDS, MS

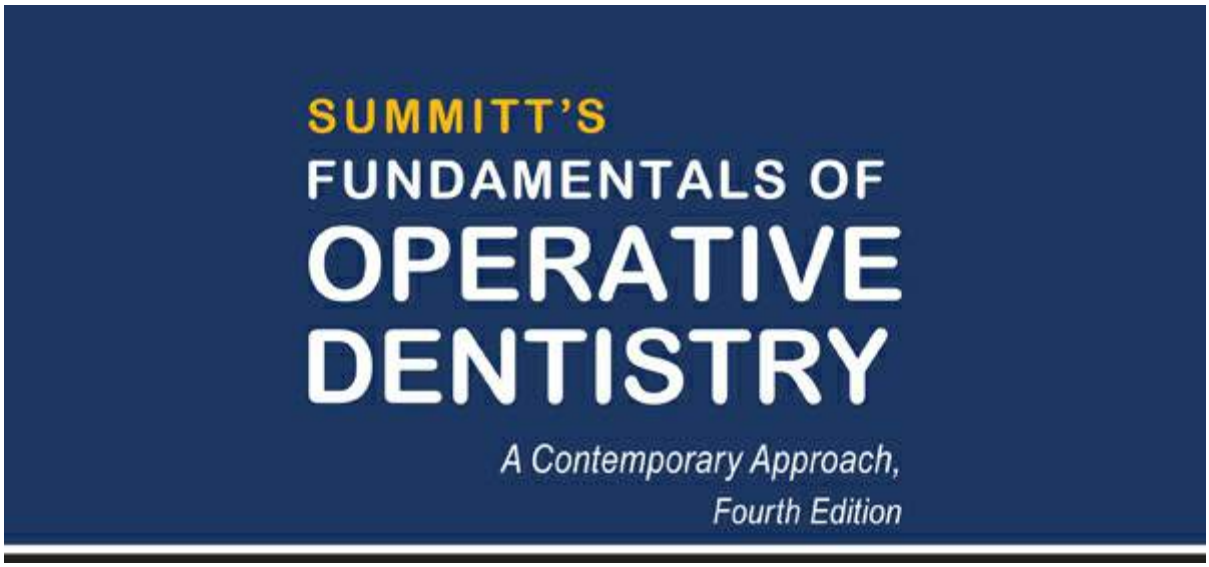
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Fourth Edition



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To my wife and best friend, DeaDea, for her constant love, support, and encouragement; to my parents for instilling in me the qualities that have served me throughout my life; and to my role models and mentors, including the coeditors and book's new namesake, for inspiring me to strive for excellence.
—TJH

To Tricia, my wife and best friend, whose support in all things means more to me than she will ever know.
—JLF

To my beautiful granddaughter, Sophia McAlister Ward, and the women who made me a grandfather—my wife and true love, Mary, and my wonderful daughter, Kristy.
—JCB

In Memoriam



Dr J. D. (Dave) Overton, DDS (1953–2013)

Dr J. D. (Dave) Overton, dds, an author of two chapters in this book, passed away on April 20, 2013, at the age of 60 years following a 4-year battle with mesothelioma. Dave was a superb operative dentistry educator, and, for 10 years, he was the leader of the faculty who taught operative dentistry at the Dental School at the University of Texas Health Science Center in San Antonio. He was unwavering in his devotion to student learning and dedicated to ensuring that the operative dentistry curriculum at his institution was based on the best evidence. That dedication is reflected in his contributions to this textbook. He will be greatly missed. His legacy of excellence in operative dentistry will benefit present and future generations of dental professionals and their patients.

Contents

In Memoriam
Preface
Contributors

1 Biologic Considerations
Terry J. Fruits, Sharukh S. Khajotia, and Jerry W. Nicholson

2 Patient Evaluation and Problem-Oriented Treatment Planning
William F. Rose, Jr, Carl W. Haveman, and Richard D. Davis

3 Esthetic Considerations in Diagnosis and Treatment Planning
J. William Robbins

4 Color and Shade Matching
Rade D. Paravina

5 Caries Management: Diagnosis and Treatment Strategies
Bennett T. Amaechi, J. Peter van Amerongen, Cor van Loveren, and Edwina A. M. Kidd

6 **Pulpal Considerations**
Thomas J. Hilton and James B. Summitt

7 **Nomenclature and Instrumentation**
James B. Summitt

8 **Field Isolation**
James B. Summitt

9 **Adhesion to Enamel and Dentin**
Lorenzo Breschi, Jack L. Ferracane, Milena Cadenaro, Annalisa Mazzoni, and Thomas J. Hilton

10 **Direct Anterior Restorations**
Marcos A. Vargas, Cathia Bergeron, David F. Murchison, Joost Roeters, and Daniel C. N. Chan

11 **Direct Posterior Esthetic Restorations**
Thomas J. Hilton and James C. Broome

12 **Amalgam Restorations**
J. D. Overton, James B. Summitt, and John W. Osborne

13 **Diagnosis and Treatment of Root Caries**
Bruce A. Matis, Carlos González-Cabezas, and Michael A. Cochran

14 **Fluoride-Releasing Materials**
Deniz Cakir-Ustun, Nathaniel C. Lawson, and John O. Burgess

15 **Class 5 Restorations**
J. D. Overton, Thomas J. Hilton, Mark L. LittleStar, and Clifford B. Starr

16 **Natural Tooth Bleaching**
Van B. Haywood, Juliana da Costa, and Thomas G. Berry

17 **Porcelain Veneers**
Jeffrey S. Rouse and J. William Robbins

18 **Anterior Ceramic Crowns**
Jeffrey S. Rouse

19 **Esthetic Inlays and Onlays**
Dennis J. Fasbinder, Gisele Neiva, and J. William Robbins

20 **Cast Gold Restorations**
Patrice P. Fan, Richard Stevenson, and Thomas G. Berry

21 **Restoration of Endodontically Treated Teeth**
James C. Broome and J. William Robbins

Preface

This textbook is about contemporary operative dentistry. The Academy of Operative Dentistry has defined *operative dentistry* as that branch of dentistry concerned with the management of teeth, by direct or indirect means, that are defective through disease, trauma, wear, and/or abnormal development, or are unesthetic, to a state of normal form, function, health, and appearance. This includes preventive/preservative, diagnostic, biologic, mechanical, and therapeutic procedures, applying all relevant aspects of dental technology and biomaterials and other oral and dental sciences.

The book is a blend of traditional, time-proven methods and recent scientific developments. Whereas preparations for cast gold restorations have changed relatively little over the years, preparations for amalgam and resin composite restorations are smaller and allow for less sound tooth structure to be removed because of the development of adhesive technologies. While we still use many luting agents in the traditional manner, adhesive cements provide greater retention for cast restorations and allow expanded use of ceramic and resin composite materials. Many concepts of caries management and pulpal protection have changed drastically as well. It is our hope that this textbook, which represents an ardent effort to present current concepts and the latest scientific evidence in restorative and preventive dentistry, will be helpful to students, educators, and practicing dentists during this time of rapidly developing technologies.

Several themes echo throughout this textbook. The first is the attempt to provide a scientific basis for the concepts described. The authors are clinically active, and many are engaged in clinical and laboratory research in the areas of cariology, restorative dentistry, and/or dental materials. Whenever possible, the diagnosis and treatment options described are based on current research findings. When convincing evidence is not available, we have attempted to present a consensus founded on a significant depth of experience and informed thought.

A second theme reflected in the book is our commitment to conservative dentistry. The treatment modalities described involve the preservation of as much sound tooth structure as possible within the framework of the existing destruction and the

patient's expectations for esthetic results. When disease necessitates a restoration, it should be kept as small as possible. However, it must be kept in mind that a conservative philosophy is also based on predictability. The treatment that is most predictable in terms of functional and esthetic longevity, based as much as possible on scientific evidence, must also be considered the most conservative. Therefore, when an extensive amount of tooth structure has been destroyed and remaining cusps are significantly weakened, occlusal coverage with a restoration may be the most predictable and therefore most conservative treatment. When portions of axial tooth surfaces are healthy, their preservation is desirable. In the conservative philosophy on which this book is based, a complete-coverage restoration (complete crown) is generally considered the least desirable treatment alternative, unless the tooth condition is such that a complete-coverage restoration will provide the most predictable clinical outcome.

The book describes techniques for the restoration of health, function, and esthetics of individual teeth and the dentition as a whole. Included are descriptions of direct conservative restorations fabricated from dental amalgam, resin composite, and resin-ionomer materials. Also detailed are techniques for partial- and complete-coverage indirect restorations of gold alloy, ceramics, metal-ceramic, and resin composite.

This fourth edition has been updated with new information based on evidence reported since the third edition. Because of new evidence, all chapters were revised, reference lists were expanded, and new authors were added to 11 chapters. A new chapter on color and shade matching has been added because of the increased emphasis on esthetic procedures in restorative dentistry. In addition, the chapter on adhesion to enamel and dentin has been completely rewritten.

This edition has also undergone a change in editorship with Tom Hilton taking the role of lead editor and the addition of Jack Ferracane and Jim Broome as co-editors, both of whom participated in the planning, editing, and revision of this textbook as a whole and were invaluable and tireless in seeing this project through.

As in the previous editions, the primary objective in producing this book is to provide students and practitioners with current and practical concepts of prevention and management of caries as a disease and of restoration of individual teeth. It is our hope that the changes made in this edition will make it of greater benefit to those who use it.



Title Change to this Edition

One of the significant changes to the fourth edition of the textbook is the title, with the addition of "Summitt's" before *Fundamentals of Operative Dentistry: A Contemporary Approach*. This addition is to honor Dr James B. Summitt, long-time clinician, educator, and researcher and the lead editor of the previous two editions of this book. He has been and continues to be a mentor to virtually all of the editors and authors of this textbook, as well as to many in the profession. Earlier versions of the preface have noted that Dr G. V. Black, the father of operative dentistry, was one of dentistry's greatest innovators and original thinkers. Dr Summitt is of the same breed as G. V. Black. While trained in traditional operative techniques and materials, Dr Summitt has always led the advance of new technology and innovation. Dr Summitt embodies the essence of what this textbook is about: looking to recent scientific innovations and incorporating them into our practices and dental school curricula. We are humbled and honored to know Dr Summitt as a personal friend, advocate for the profession, relentless devotee to evidence-based dentistry, and, most importantly, a role model of character and integrity.

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